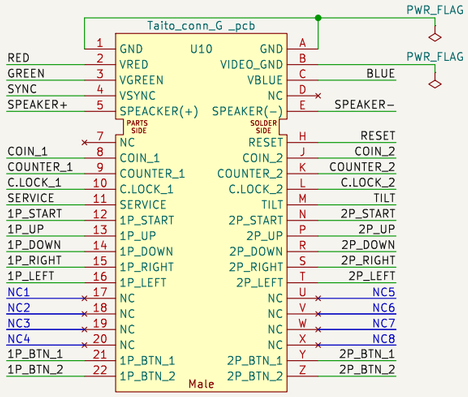
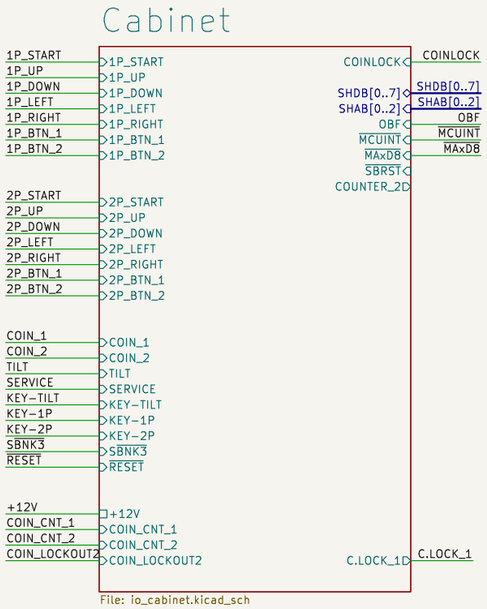
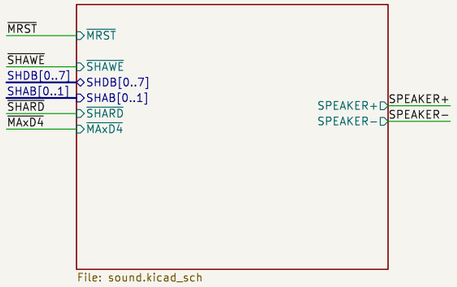
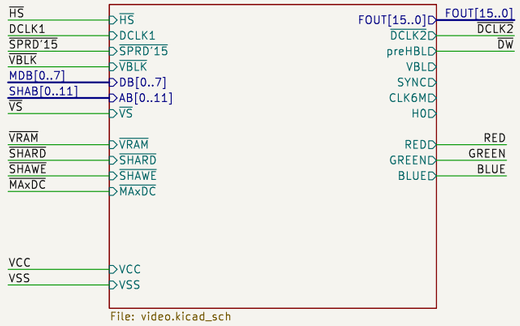
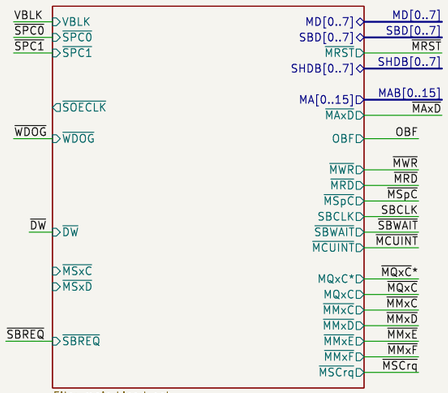


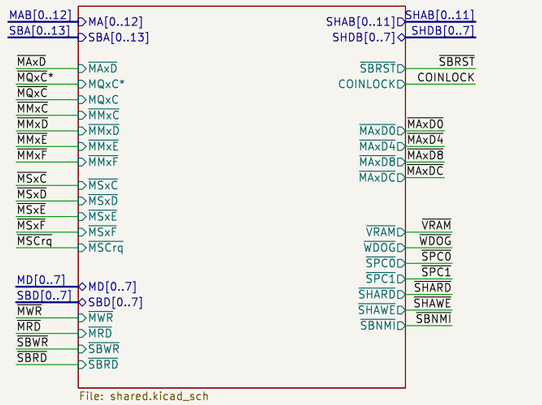


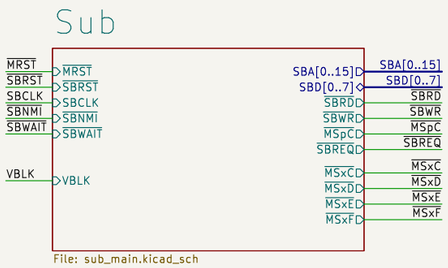


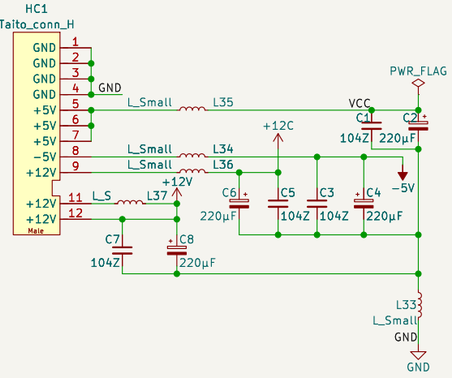


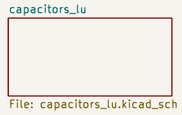










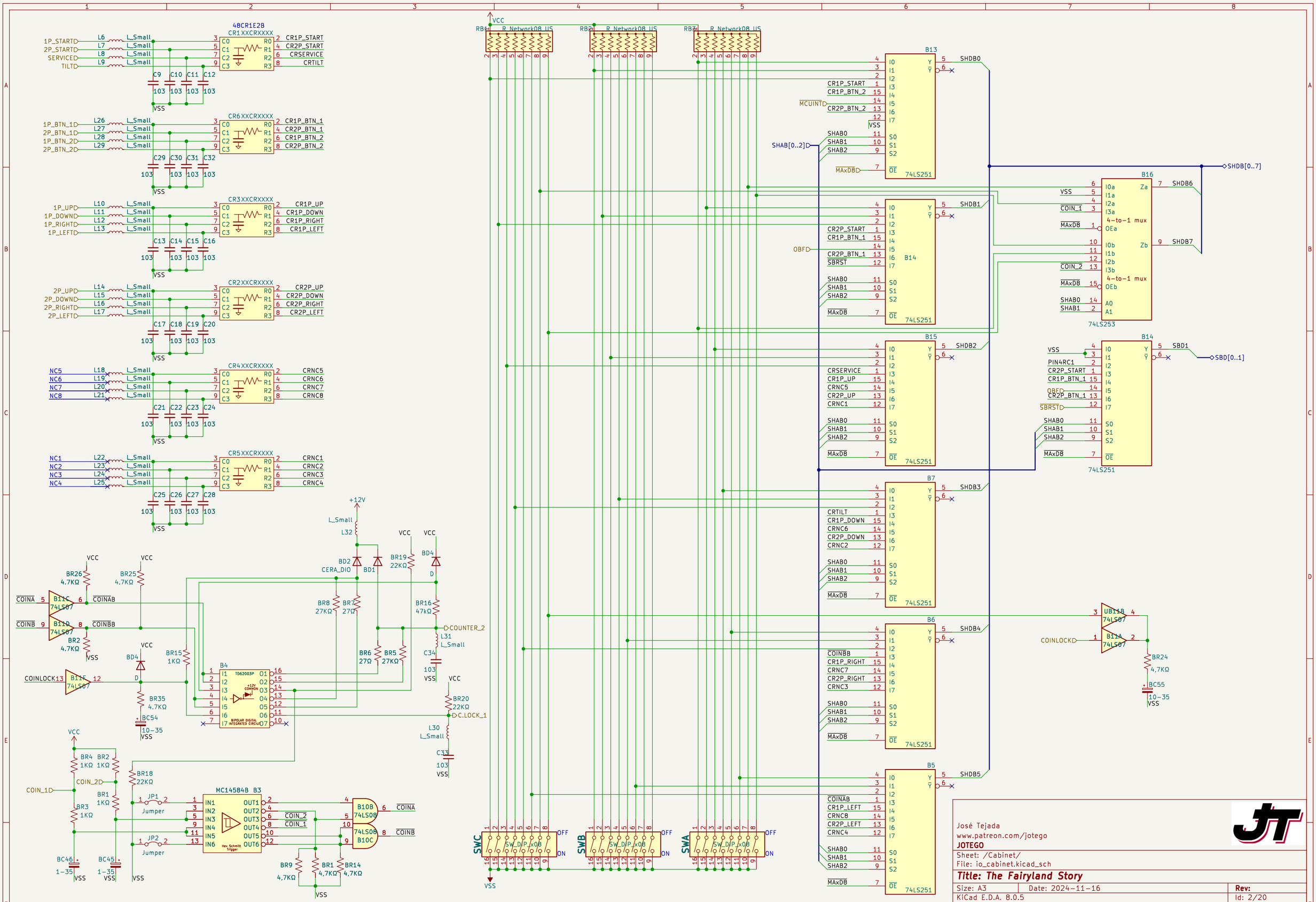


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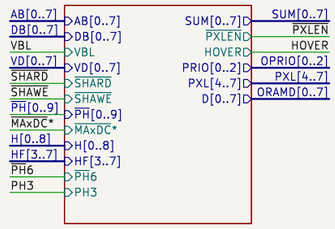
Sheet: /
File: fairyland.kicad_sch

Title: The Fairyland Story

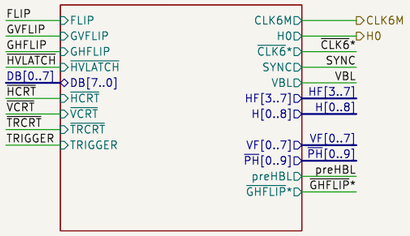
Size: A3	Date: 2024−11−16	**Rev:**
KiCad E.D.A. 8.0.5		Id: 1/20


48CR1E2B
CR1XXCRXXXX
1P_START
2P_START
SERVICE
TILT
CR1P_START
CR2P_START
CRSERVICE
CRTILT
CR6XXCRXXXX
1P_BTN_1
2P_BTN_1
1P_BTN_2
2P_BTN_2
CR1P_BTN_1
CR2P_BTN_1
CR1P_BTN_2
CR2P_BTN_2
CR3XXCRXXXX
1P_UP
1P_DOWN
1P_RIGHT
1P_LEFT
CR1P_UP
CR1P_DOWN
CR1P_RIGHT
CR1P_LEFT
CR2XXCRXXXX
2P_UP
2P_DOWN
2P_RIGHT
2P_LEFT
CR2P_UP
CR2P_DOWN
CR2P_RIGHT
CR2P_LEFT
CR4XXCRXXXX
NC5
NC6
NC7
NC8
CRNC5
CRNC6
CRNC7
CRNC8
CR5XXCRXXXX
NC1
NC2
NC3
NC4
CRNC1
CRNC2
CRNC3
CRNC4
VSS
VCC
+12V
R_Network08_US
RB1
RB2
RB3
B13
B14
B15
B7
B6
B5
74LS251
SHAB[0..2]
MCUINT
OBF
SBRST
MAxD8
SHDB0
SHDB1
SHDB2
SHDB3
SHDB4
SHDB5
SHDB[0..7]
B16
74LS253
4-to-1 mux
SBD1
SBD[0..1]
PIN4RC1
COIN_1
COIN_2
B11C
B11D
B11E
B11A
B11B
74LS07
COINA
COINB
COINLOCK
COINAB
COINBB
COINLOCKD
B4
TD62003P
BIPOLAR DIGITAL INTEGRATED CIRCUIT
MC14584B B3
Hex Schmitt Trigger
B10B
B10C
74LS08
JP1
JP2
Jumper
COUNTER_2
C.LOCK_1
CERA_DIO
SWC
SWB
SWA
SW_DIP_X08
OFF
ON
BR1
BR2
BR3
BR4
BR5
BR6
BR7
BR8
BR9
BR14
BR15
BR16
BR18
BR19
BR20
BR24
BR25
BR26
BR35
BC45
BC46
BC54
BC55
BD1
BD2
BD4
C34
C33
L30
L31
L32
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JOTEGO
Sheet: /Cabinet/
File: io_cabinet.kicad_sch
Title: The Fairyland Story
Size: A3
Date: 2024-11-16
KiCad E.D.A. 8.0.5
Rev:
Id: 2/20

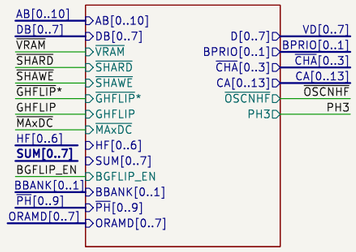
Object RAM



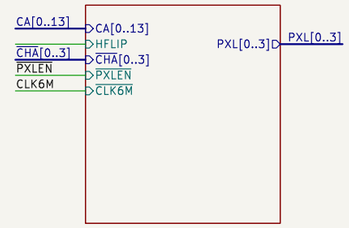
Timing



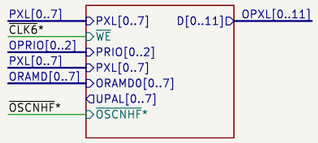
Video RAM



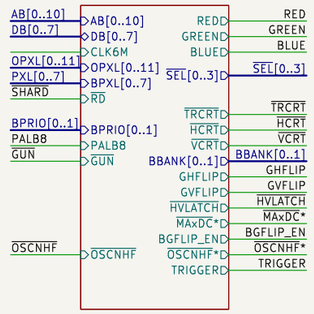
Tile ROM



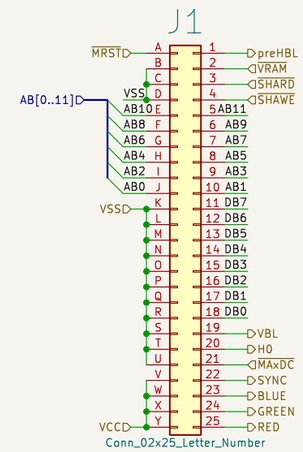
Line buffer

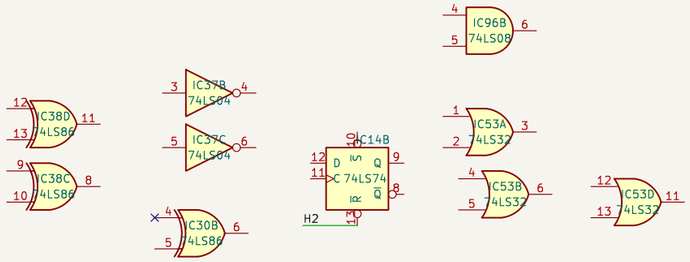


Palette



J1



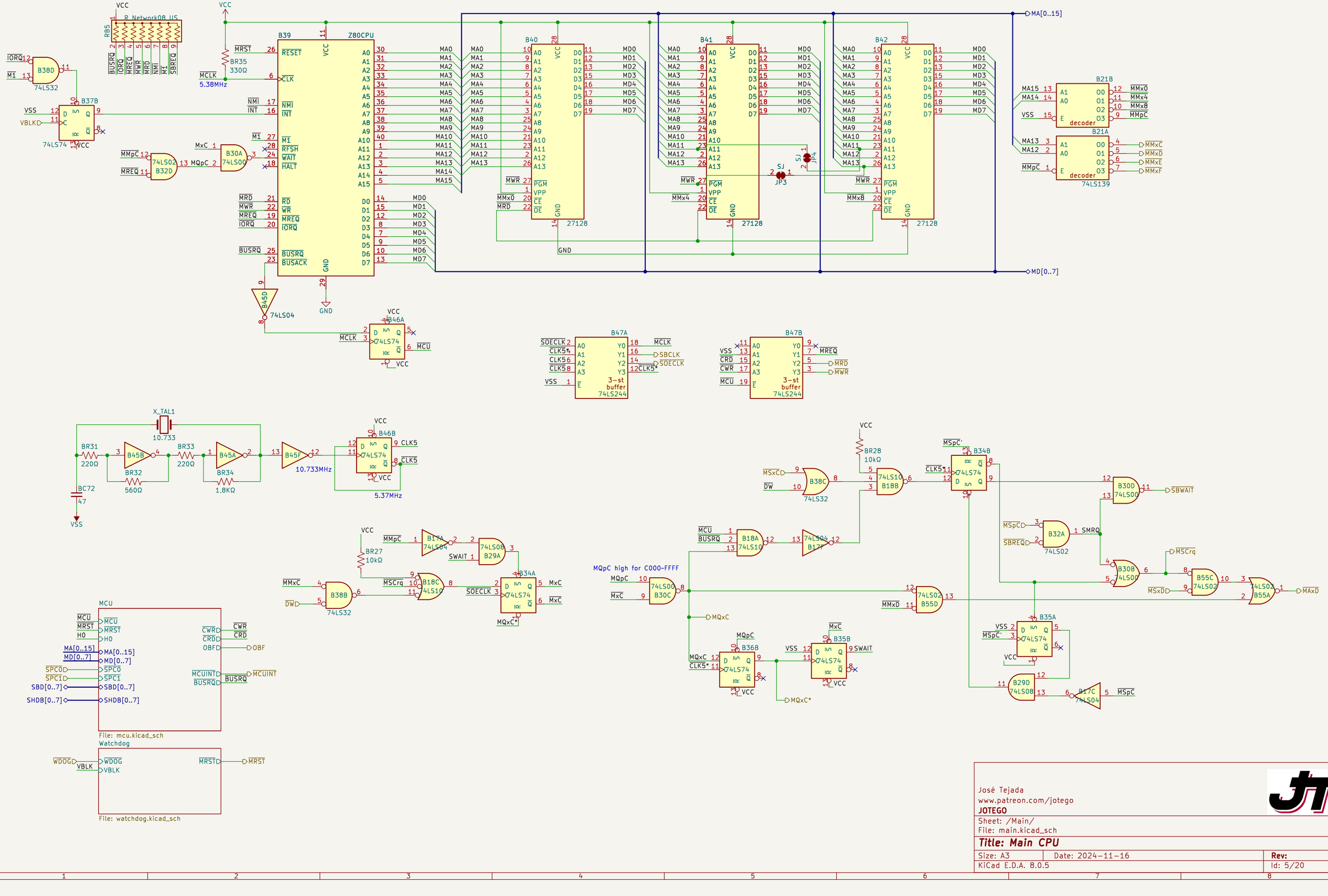


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Sheet: /Video/
File: video.kicad_sch

Title: The Fairyland Story

Size: A3	Date: 2024-11-16	**Rev:**
KiCad E.D.A. 8.0.5		Id: 4/20


VCC
R_Network08_US
RB5
BUSRQ
IORQ
MREQ
MWR
MRD
NMI
M1
SBREQ
IORQ
M1
B38D
74LS32
B37B
VSS
VBLK
74LS74
VCC
MMpC
MREQ
74LS02
B32D
MQpC
MxC
B30A
74LS00
MCLK
5.38MHz
BR35
330Ω
B39
Z80CPU
RESET
CLK
NMI
INT
M1
RFSH
WAIT
HALT
MRD
MWR
MREQ
IORQ
BUSRQ
BUSACK
GND
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
A13
A14
A15
D0
D1
D2
D3
D4
D5
D6
D7
MRST
MA0
MA1
MA2
MA3
MA4
MA5
MA6
MA7
MA8
MA9
MA10
MA11
MA12
MA13
MA14
MA15
MD0
MD1
MD2
MD3
MD4
MD5
MD6
MD7
B40
B41
B42
27128
PGM
VPP
CE
OE
MMx0
MMx4
MMx8
SJ
JP4
JP3
MA[0..15]
MD[0..7]
B21B
B21A
decoder
74LS139
MMx0
MMx4
MMx8
MMpC
MMxC
MMxD
MMxE
MMxF
B45D
74LS04
B46A
74LS74
MCU
B47A
B47B
74LS244
SOECLK
CLK5*
CLK5
SBCLK
CRD
CWR
MCU
MREQ
MRD
MWR
X_TAL1
10.733
BR31
220Ω
BR32
560Ω
BR33
220Ω
BR34
1.8KΩ
BC72
47
B45B
B45A
B45F
10.733MHz
B46B
5.37MHz
BR27
10kΩ
B17A
B38B
B18C
B34A
B29A
SWAIT
MSCrq
MAxD
BR28
10kΩ
B38C
B18B
74LS10
B18A
B17B
74LS04
MSxCD
DW
MCU
BUSRQ
MQpC high for C000-FFFF
MQpC
MxC
B30C
MQxC
MQxC*
B36B
B35B
B34B
MSpC'
B30D
B32A
SMRQ
SBREQ
SBWAIT
B30B
B55C
B55D
B55A
MSxDD
B35A
B29D
B17C
MSpC
MCU
MRST
H0
MA[0..15]
MD[0..7]
SPC0
SPC1
SBD[0..7]
SHDB[0..7]
CWR
CRD
OBF
MCUINT
BUSRQ
File: mcu.kicad_sch
Watchdog
WDOG
VBLK
File: watchdog.kicad_sch
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Sheet: /Main/
File: main.kicad_sch
Title: Main CPU
Size: A3
Date: 2024-11-16
Rev:
KiCad E.D.A. 8.0.5
Id: 5/20

Clocks

File: snd_clocks.kicad_sch

Sound Generators

File: ongen.kicad_sch

Latches

File: latches.kicad_sch

Analog

File: amplifiers.kicad_sch

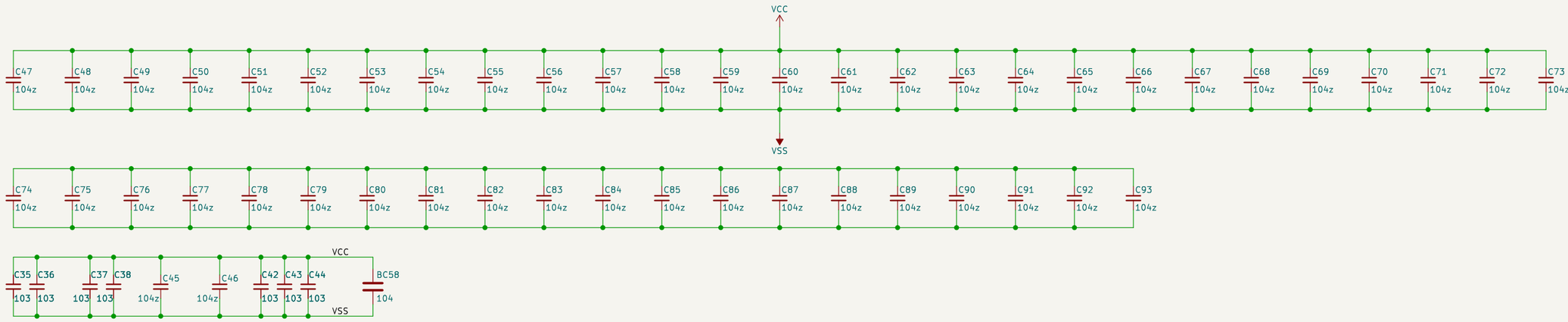
NOT POPULATED

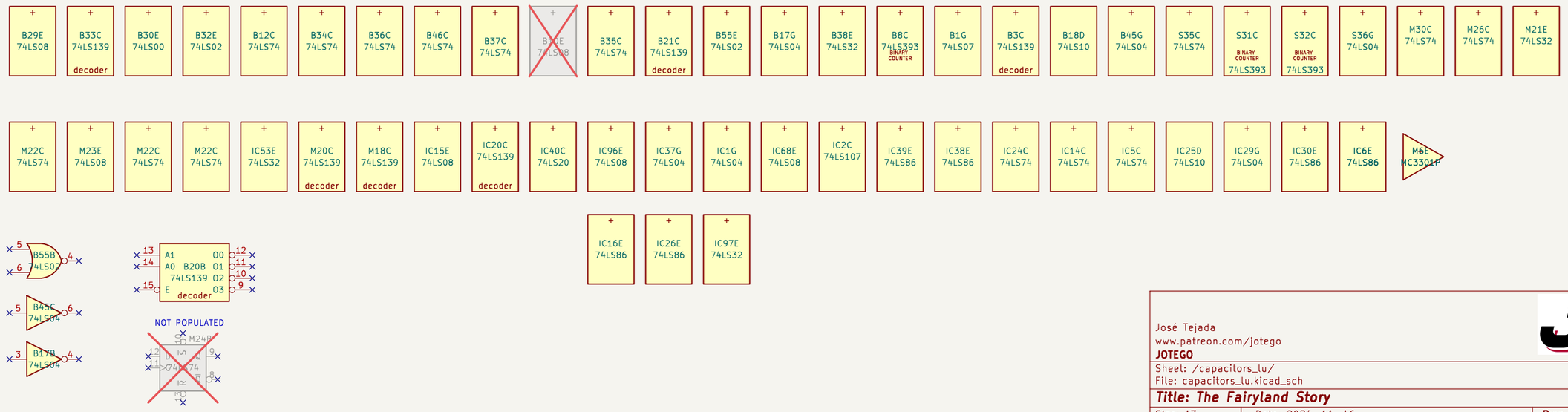
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Sheet: /Sound/
File: sound.kicad_sch

Title: Sound Board

Size: A3	Date: 2024-11-16	**Rev:**
KiCad E.D.A. 8.0.5		Id: 7/20



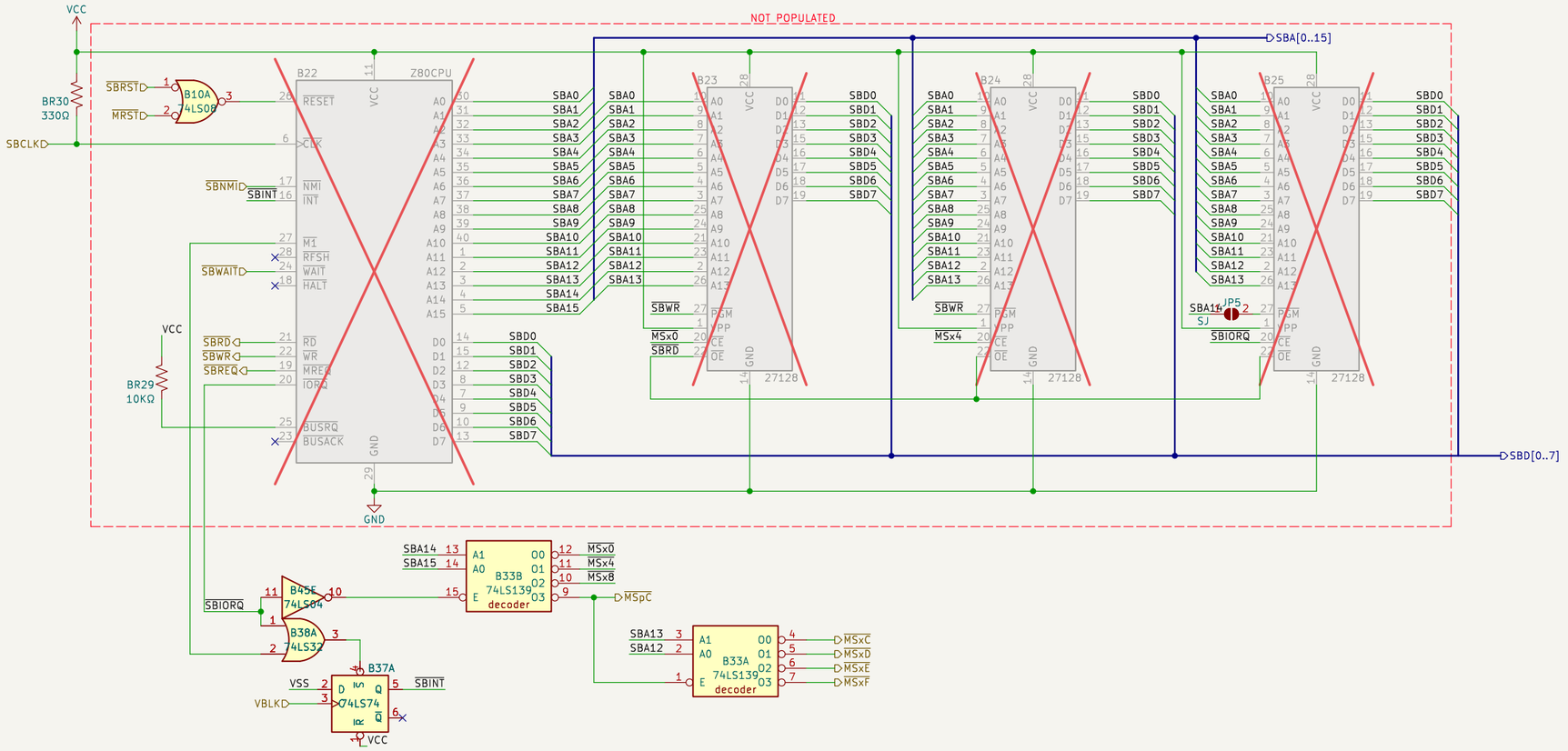


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Sheet: /capacitors_lu/
File: capacitors_lu.kicad_sch

Title: The Fairyland Story

Size: A3	Date: 2024-11-16	**Rev:**
KiCad E.D.A. 8.0.5		Id: 8/20



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Sheet: /Sub/
File: sub_main.kicad_sch

Title: Subsidiary CPU

Size: A3	Date: 2024-11-16	**Rev:**
KiCad E.D.A. 8.0.5		Id: 10/20

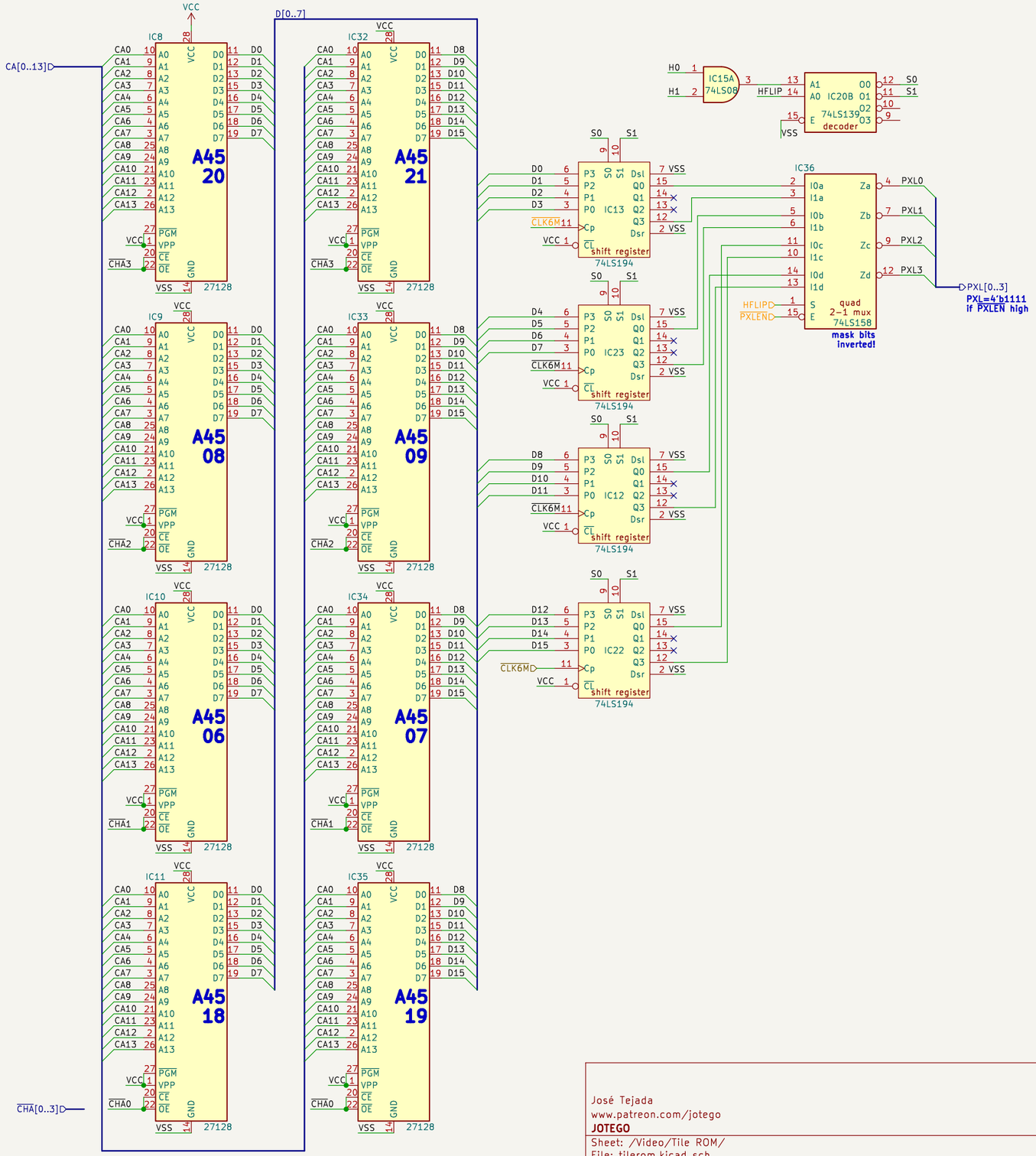
PULL-D7
MC40 472K
MR33 1MΩ
MR32 20MΩ
MR82 1.5kΩ
VSS
M4B MC3301P
DA TEST
MR34 20MΩ
MC41 151K
MC39 104K
MR38 4.7kΩ
+12V
MC18 332K
MC19 473K
MC20 104K
MC21 104K
MC79 104K
MC81 104K
MR11 1.5kΩ
MC24 47µF
M4E MC3301P
MR83 10kΩ
MR36 5kΩ
MC68 10µF
MR37 1MΩ
MC16 47µF
MR6 100kΩ
MR14 10kΩ
MR16 10kΩ
MR10 1.5kΩ
M2
TA7630P
REF_SIG TH(2) TL(2) TL(1) TH(1) VCC
IN-2 IN-1 OUT-2 OUT-1
Vref Balance Volume Bass Treble VSS
DUAL VOLUME/BALANCE /TONE(BASS/TREBLE)
PSG
MR8 10kΩ
MC17 104K
MR9 1kΩ
MR7 10kΩ
MR15 75kΩ
MR17 75kΩ
MC26 104K
MR85 1MΩ
OUT-1
M4D MC3301P
MR86 1MΩ
MC69 3.3
MR40 1MΩ
TEST KG
MC15 10µF
MR35 4.7kΩ
MR84 1MΩ
M4A MC3301P
MC70 3.3
MR39 1MΩ
M4C MC3301P
MR41 1MΩ
UK TEST? M12
NOT POPULATED
MR2 7.6kΩ
MR3 1kΩ
MC15 472K
VR1 R_P_US
MC1 10µF
VCC_AMP
MC38 104Z
MC4 220µF
M1 SB3731
IN FEEDBACK BYPASS GND-I GND-O MUTE BS-I OUT-I N.C VCC BS-M OUT-M
MC2 10µF
MR1 100kΩ
MC3 220µF
MC10 100µF
MC11 104K
MC12 100µF
MC13 154J
MC14 104K
MR4 10Ω
MR5 10Ω
SPEAKER+
SPEAKER-
AMUTED
+12V
MR13 1.5kΩ
7.7V
MC83 104K
MC25 47µF
MR12 2.7kΩ
MR48 1MΩ
CTLREF
MR49 560kΩ
PTREBLE
M5C MC3301P
MC74 472K
MR52 1MΩ
PBALD
MR43 560kΩ
CTLREF
MR42 1MΩ
M5B MC3301P
MR47 1MΩ
MC71 472K
MC72 472K
MR46 1MΩ
M5A MC3301P
CTLREF
MR44 1MΩ
PVOL
MR45 560kΩ
MC78 472K
MR53 1MΩ
M5D MC3301P
MR50 1MΩ
CTLREF
MR51 560kΩ
PBASS
For the PSG
Each analog control voltage goes
from 0.55V to 3.7V at the opamp output
Compared to the reference, that makes
from −7.15V (0V at DAC) to −4V (5V at DAC)
MC30 332K
MC31 473K
MC32 104K
MC33 104K
MC80 104K
MR25 1.5kΩ
MC34 47µF
M5E MC3301P
MC36 10µF
MR23 6.8kΩ
1.54V CTLREF
MR22 1kΩ
MR19 100kΩ
IN-2
IN-1
MR21 100kΩ
MC27 47µF
M3 MR28 10kΩ
MR30 10kΩ
MR24 1.5kΩ
TONEB
TONEA
MR18 1kΩ
MC29 104K
MR20 1kΩ
MC28 104K
MR29 75kΩ
MR31 75kΩ
104K
OUT-1
MC37
TA7630P
+12V
MR27 1.5kΩ
7.7V
MC82 104K
MC35 47µF
MR26 2.7kΩ
BALD
MR57 620kΩ
CTLREF
MR56 1MΩ
M6A MC3301P
MR58 1MΩ
MC77 472K
M6D MC3301P
MR62 1MΩ
CTLREF
MR63 620kΩ
TREBLE
M6C MC3301P
MR60 1MΩ
MR61 620kΩ
BASS
M6B MC3301P
MR54 1MΩ
MR55 620kΩ
VOL
For the MSM
Each analog control voltage goes
from 0.60V to 3.7V at the opamp output
Compared to the reference, that makes
from −7.1V (0V at DAC) to −4V (5V at DAC)
SHAWED
JP2 SJ
PIN 27 SOCKET
NOT USED - B49
SC1
MRST MAxD4 SHARD
SHDB[0..7]
SHDB7 SHDB6 SHDB5 SHDB4 SHDB3 SHDB2 SHDB1 SHDB0
SHAB[0..1]
SHAB1 SHAB0
VSS AVSS VCC PWR_FLAG
SPEAKER+ SPEAKER-
L40 L_Small
L39 L_Small
C39 103
C40 103
NO MOUNTED
Conn_01x10_Pin
MN1
+12V
VCC
L38 L_Small
C6 104Z
MC7 220µF
C8 104Z
MC9 220µF
PWR_FLAG
VCC_AMP
AVSS
VCC

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Sheet: /Sound/Analog/
File: amplifiers.kicad_sch

Title: Analog Signal Processing

Size: A3	Date: 2024-11-16	Rev:
KiCad E.D.A. 8.0.5		Id: 11/20



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Sheet: /Video/Tile ROM/
File: tilerom.kicad_sch

Title: Tile ROMs

Size: A3	Date: 2024-11-16	Rev:
KiCad E.D.A. 8.0.5		Id: 12/20

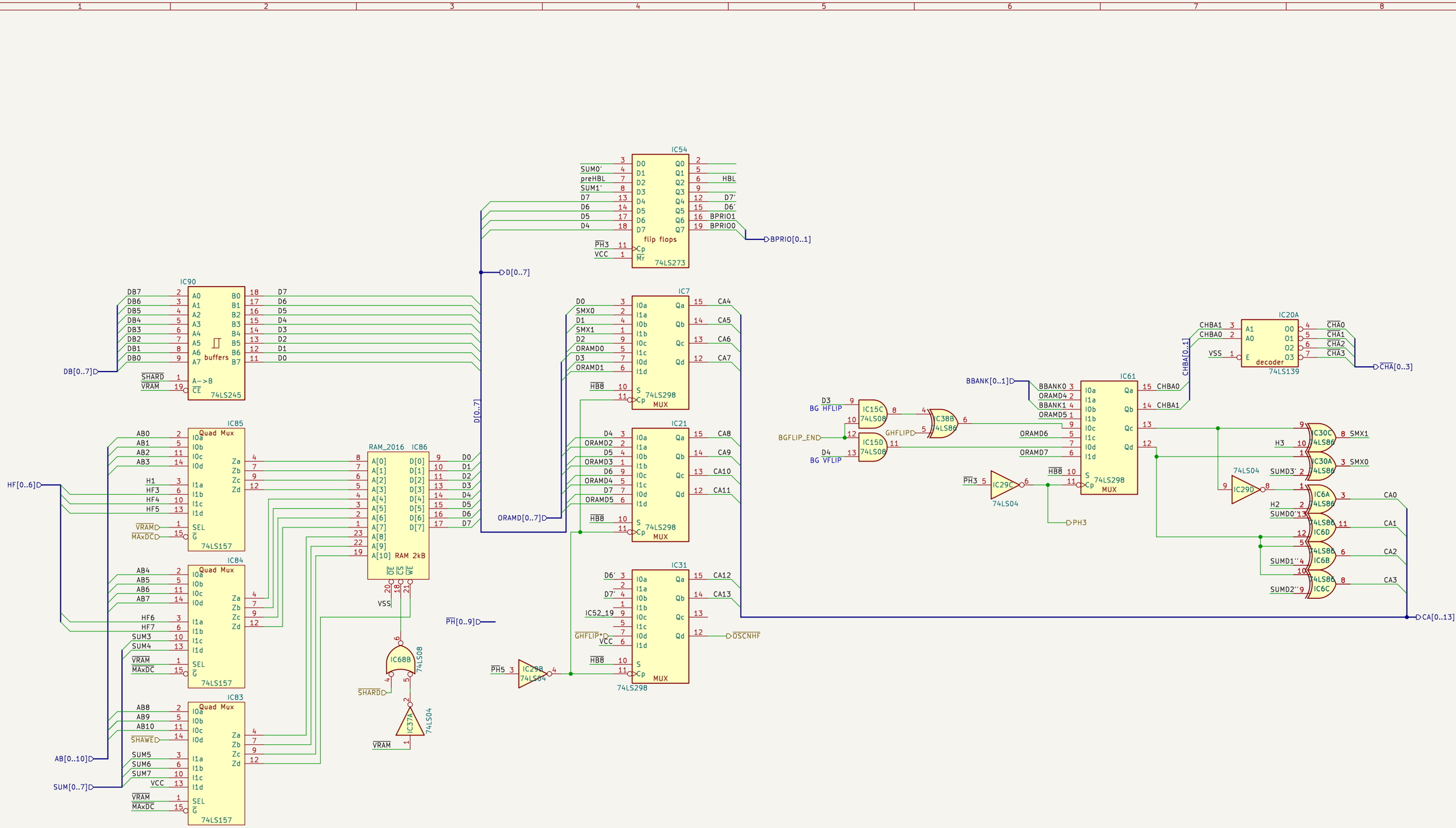
XTAL1
18.432MHz
BC6
20
IC1D
74LS04
SC7
20
IC1F
74LS04
BR2
330Ω
BR9
330Ω
IC1E
74LS04
IC2A
74LS107
CLK6M
CLK6*
IC2B
74LS107
6.144MHz
IC1A
74LS04
CLK6M
IC14A
74LS74
GHFLIP
PH1
PH6
GHFLIP*
IC18
74LS161
counter
IC28
74LS161
counter
IC5A
74LS74
IC5B
74LS74
IC1C
74LS04
H[0..8]
HF[3..7]
During PH6, HF=H
IC39B
IC39D
IC39C
IC39A
IC38A
IC70
4-LINE BCD TO 10-LINE DECIMAL DECODERS
74LS42
3.08MHz H0
1.54MHz H1
769kHz H2
6.25MHz CLK6M
PH[0..9]
769kHz
Each signal is active
every 8 pixels
IC29E
74LS04
IC15B
74LS08
IC19A
74LS74
HBB
IC30D
74LS86
SYNC
IC25A
74LS10
preHBL
IC25B
74LS10
IC19B
74LS74
VBL
line 240. VBL starts
line 16, VBL ends
VF[0..7]
V[0..7]
IC26C
IC16A
IC16D
IC16B
IC16C
IC26A
IC26D
IC26B
GVFLIP
IC17
74LS161
IC27
74LS161
IC1B
74LS04
HORIZONTAL COUNTER
from 080 ~ 1FF
blanking 080~0FF (128 pixels)
active 100~1FF (256 pixels)
VERTICAL COUNTER
from 0F0 ~ 1FF
blanking 1F0~1FF~0F0~0FF (32 lines)
active 100~1EF (240 lines)
R_Network08_US
DB[7..0]
IC41
74LS373
IC42
74LS373
IC52_19
VCRT
HCRT
LATCHES AND FLIP FLOPS
This seems to be
a circuit to capture
when an event occurs
relative to the raster
position. It might
have been planned for a gun.
HVLATCH
IC3A
IC3B
74LS74
TRCRT
IC4C
74LS125
TRIGGER
L2
C1
221
IC29F
74LS04

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Sheet: /Video/Timing/
File: timing.kicad_sch

Title: Video Timing

Size: A3 | Date: 2024-11-16 | **Rev:**
KiCad E.D.A. 8.0.5 | Id: 13/20



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Sheet: /Video/Video RAM/
File: vram.kicad_sch

Title: Background Video RAM

Size: A3	Date: 2024-11-16	**Rev:**
KiCad E.D.A. 8.0.5		Id: 13/20

device select signals driven by either sub or main CPUs

PIN2 SOCKET B48 (NOT USED)

Cabinet I/O

PCB can support 4kB usage too

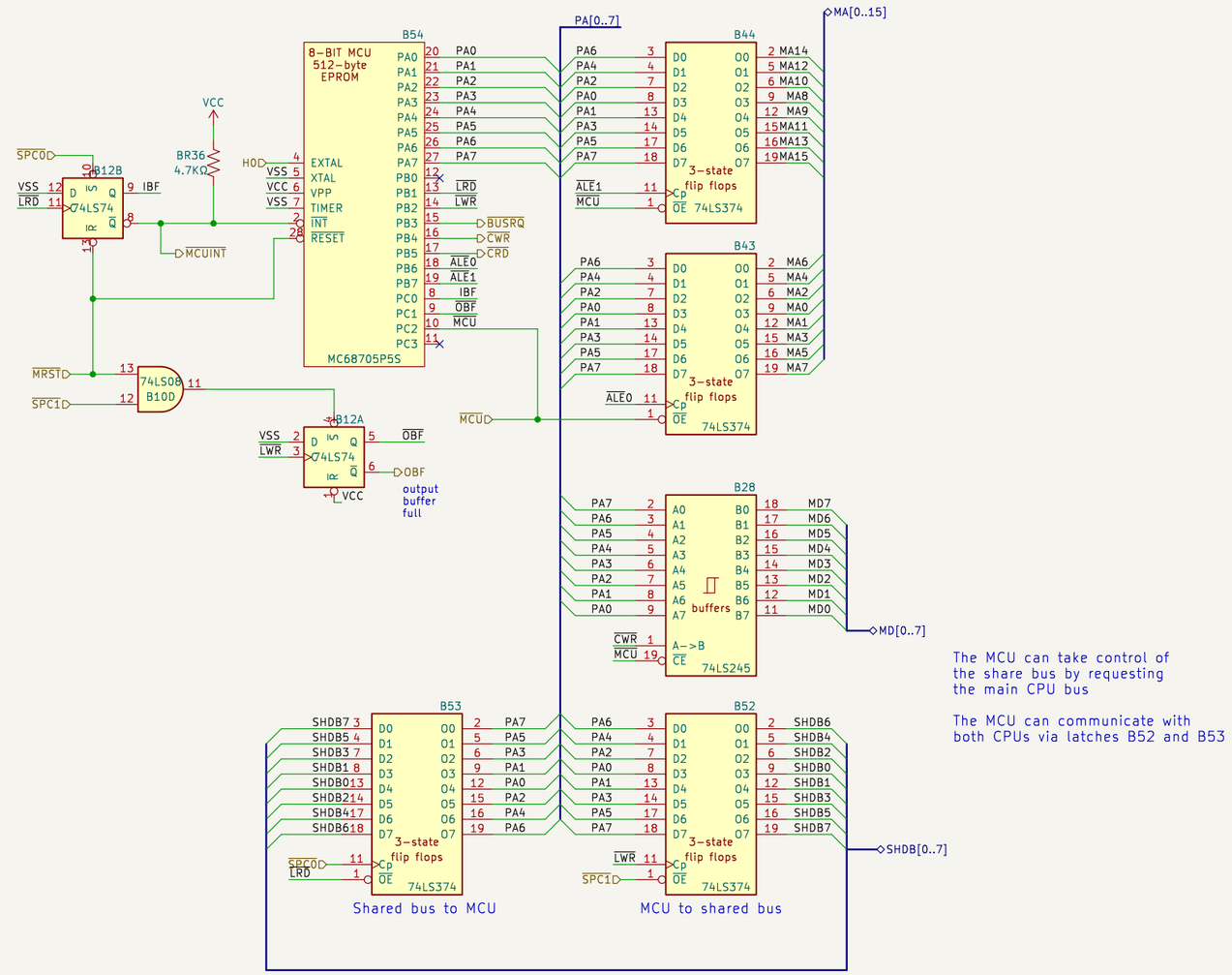
Used in NY Captor as ROM bank control

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Sheet: /Shared/
File: shared.kicad_sch

Title: Shared bus and RAM

Size: A3	Date: 2024-11-16	Rev:
KiCad E.D.A. 8.0.5		Id: 13/20

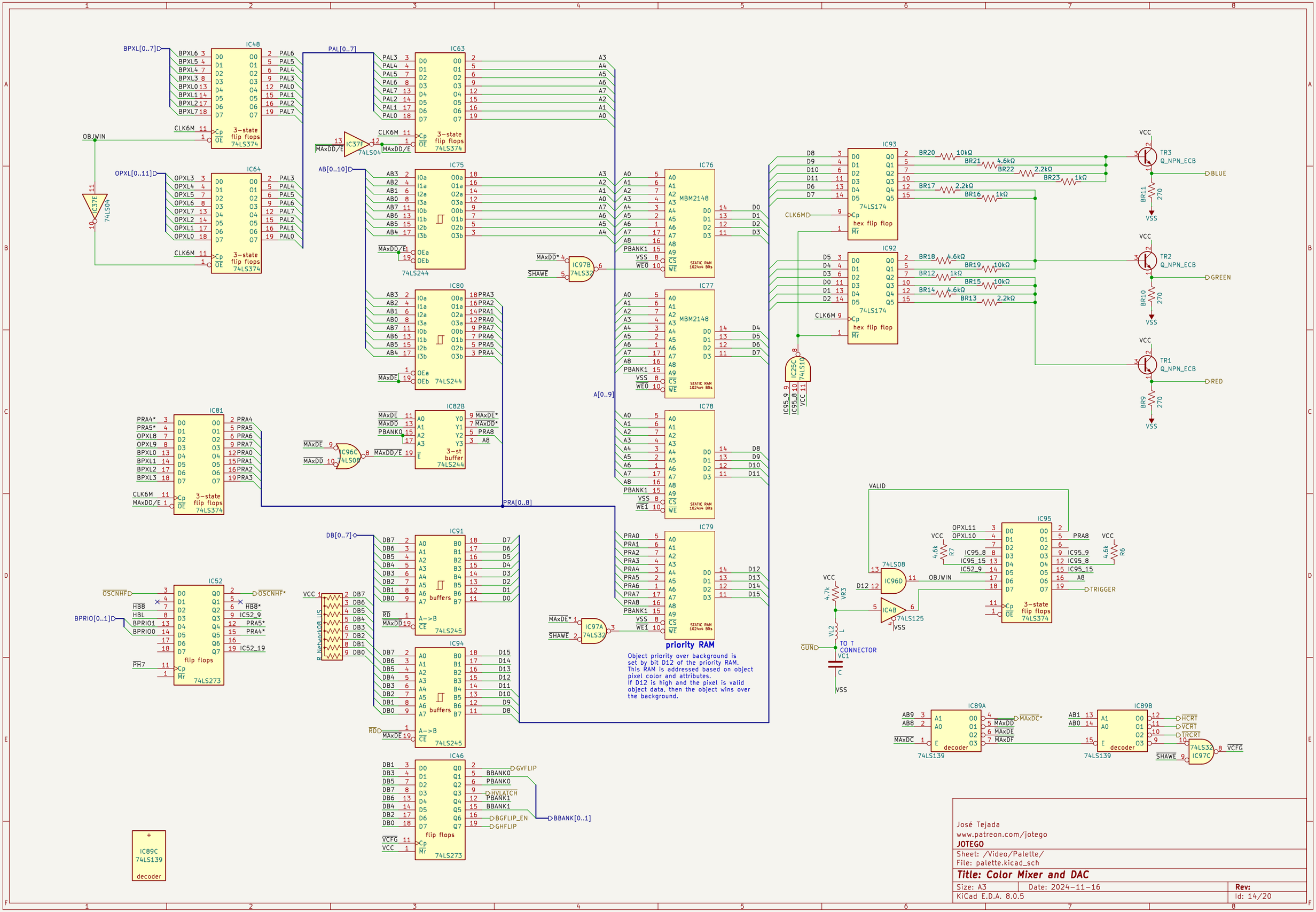


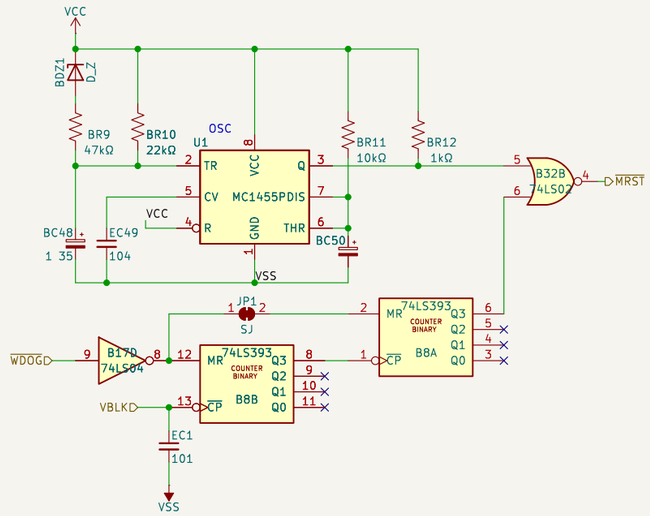
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Sheet: /Main/MCU/
File: mcu.kicad_sch

Title: MCU

Size: A3	Date: 2024-11-16	**Rev:**
KiCad E.D.A. 8.0.5		Id: 14/20


IC48
BPXL[0..7]
BPXL6 BPXL5 BPXL4 BPXL3 BPXL0 BPXL1 BPXL2 BPXL7
PAL6 PAL5 PAL4 PAL3 PAL0 PAL1 PAL2 PAL7
CLK6M
OBJWIN
3-state flip flops 74LS374
IC64
OPXL[0..11]
OPXL3 OPXL4 OPXL5 OPXL6 OPXL7 OPXL2 OPXL1 OPXL0
PAL3 PAL4 PAL5 PAL6 PAL7 PAL2 PAL1 PAL0
IC37E 74LS04
IC63
PAL[0..7]
IC37F 74LS04
MAxDD/E
IC75
AB[0..10]
AB3 AB2 AB1 AB0 AB7 AB6 AB5 AB4
A3 A4 A5 A6 A7 A2 A1 A0
MAxDD/E1
OEa OEb
74LS244
IC80
PRA3 PRA2 PRA1 PRA0 PRA7 PRA6 PRA5 PRA4
MAxDE
IC81
PRA4* PRA5* OPXL8 OPXL9 BPXL0 BPXL1 BPXL2 BPXL3
IC82B
MAxDE MAxDD PBANK0
MAxDD/E
3-st buffer 74LS244
IC52
OSCNHF
HB8 HBL BPRIO1 BPRIO0 PH7
BPRIO[0..1]
HB8* IC52_9 PRA5* PRA4* IC52_19
flip flops 74LS273
IC91
DB[0..7]
DB7 DB6 DB5 DB4 DB3 DB2 DB1 DB0
D7 D6 D5 D4 D3 D2 D1 D0
buffers
A->B
RD MAxDD
R_Network08_US
IC94
D15 D14 D13 D12 D11 D10 D9 D8
RD MAxDE
IC46
DB1 DB3 DB5 DB7 DB6 DB4 DB2 DB0
GVFLIP BBANK0 PBANK0 HVLATCH PBANK1 BBANK1 BGFLIP_EN GHFLIP
VCFG VCC
BBANK[0..1]
IC89C 74LS139 decoder
PRA[0..8]
A[0..9]
IC76
IC77
IC78
IC79
MBM2148
STATIC RAM 1024x4 Bits
MAxDD* SHAWE
IC97B 74LS32
IC97A 74LS32
D0 D1 D2 D3 D4 D5 D6 D7 D8 D9 D10 D11 D12 D13 D14 D15
priority RAM
Object priority over background is set by bit D12 of the priority RAM.
This RAM is addressed based on object pixel color and attributes.
If D12 is high and the pixel is valid object data, then the object wins over the background.
IC93
IC92
74LS174
hex flip flop
CLK6MD
CLK6M
IC25C 74LS10
IC95_9 IC95_8 VCC
BR20 10kΩ BR21 4.6kΩ BR22 2.2kΩ BR23 1kΩ BR17 2.2kΩ BR16 1kΩ
BR18 4.6kΩ BR19 10kΩ BR12 1kΩ BR15 10kΩ BR14 4.6kΩ BR13 2.2kΩ
TR3 TR2 TR1 Q_NPN_ECB
BLUE GREEN RED
BR11 BR10 BR9 270
VSS
VALID
IC95
OPXL11 OPXL10 IC95_8 IC95_15 IC52_9
PRA8 IC95_9 IC95_8 IC95_15 A8
TRIGGER
3-state flip flops 74LS374
R7 R6 4.6k
IC96D 74LS08
IC4B 74LS125
OBJWIN
VR3 4.7k
VR2
TO T CONNECTOR
VC1
GUN
IC89A
IC89B
AB9 AB8 MAxDC
MAxDD MAxDE MAxDF
MAxDC*
HCRT VCRT TRCRT
IC97C 74LS32
SHAWE
VCFG
74LS139 decoder
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Sheet: /Video/Palette/
File: palette.kicad_sch
Title: Color Mixer and DAC
Size: A3
Date: 2024-11-16
Rev:
KiCad E.D.A. 8.0.5
Id: 14/20



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Sheet: /Main/Watchdog/
File: watchdog.kicad_sch

Title:

Size: A3	Date: 2024-11-16	**Rev:**
KiCad E.D.A. 8.0.5		Id: 15/20

selects tilemap palette during active video, sprite palette otherwise

IDX[0..4] holds the sprite index to be drawn next

PXLEN
low during active video to dump the background and during H blanking only if sprite data is needed

Complex multiplexer driving
the 256-byte memory because it holds
three kinds of information
00~7F: 4-byte sprite data (32 sprites)
80~9F: sprite drawing order (bits 4:0) and sprite priority (bit 7)
A0~BF: column scroll
D0~DF: matched sprites for next line

After the first eight object matches, new ones overwrite old ones every other frame

Count down from 7, during active H each time a matching for next line object is found

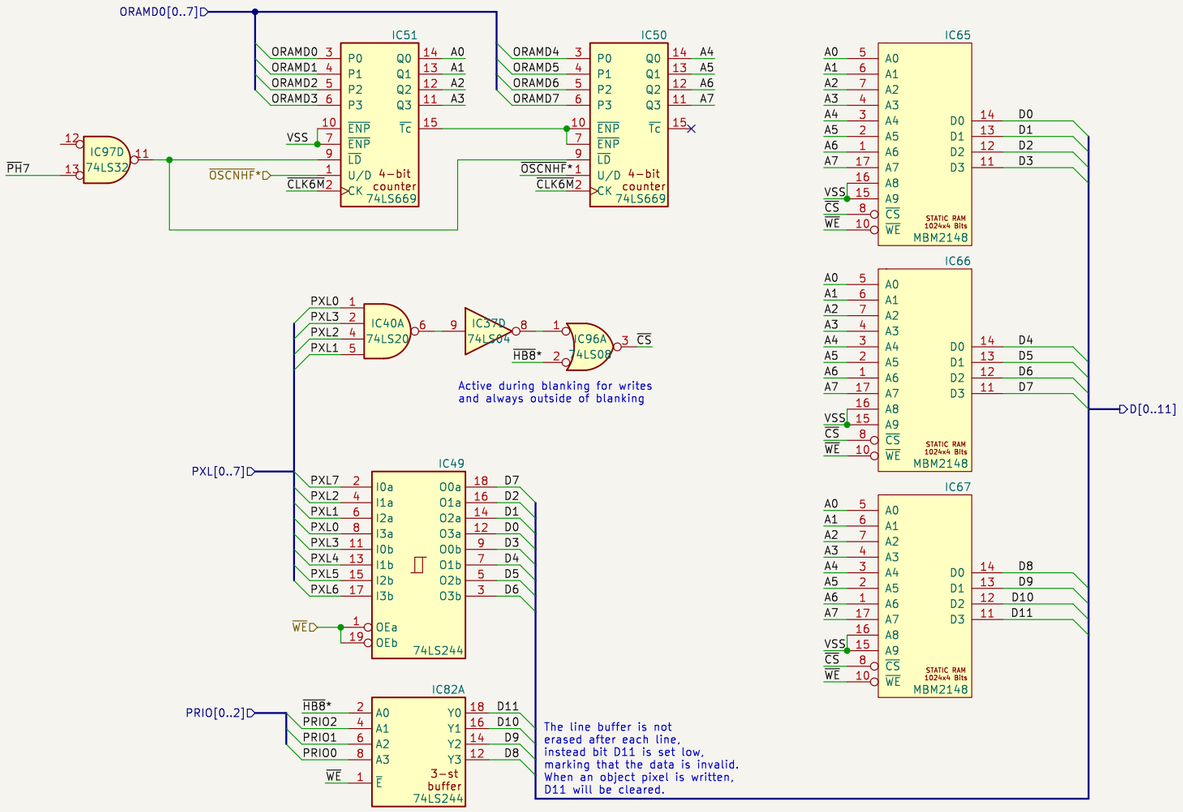
MAxDC* LOW -> CPU access
HIGH -> GFX access

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Sheet: /Video/Object RAM/
File: obj_ram.kicad_sch

Title: Object Information

Size: A3 | Date: 2024-11-16 | **Rev:**
KiCad E.D.A. 8.0.5 | Id: 16/20

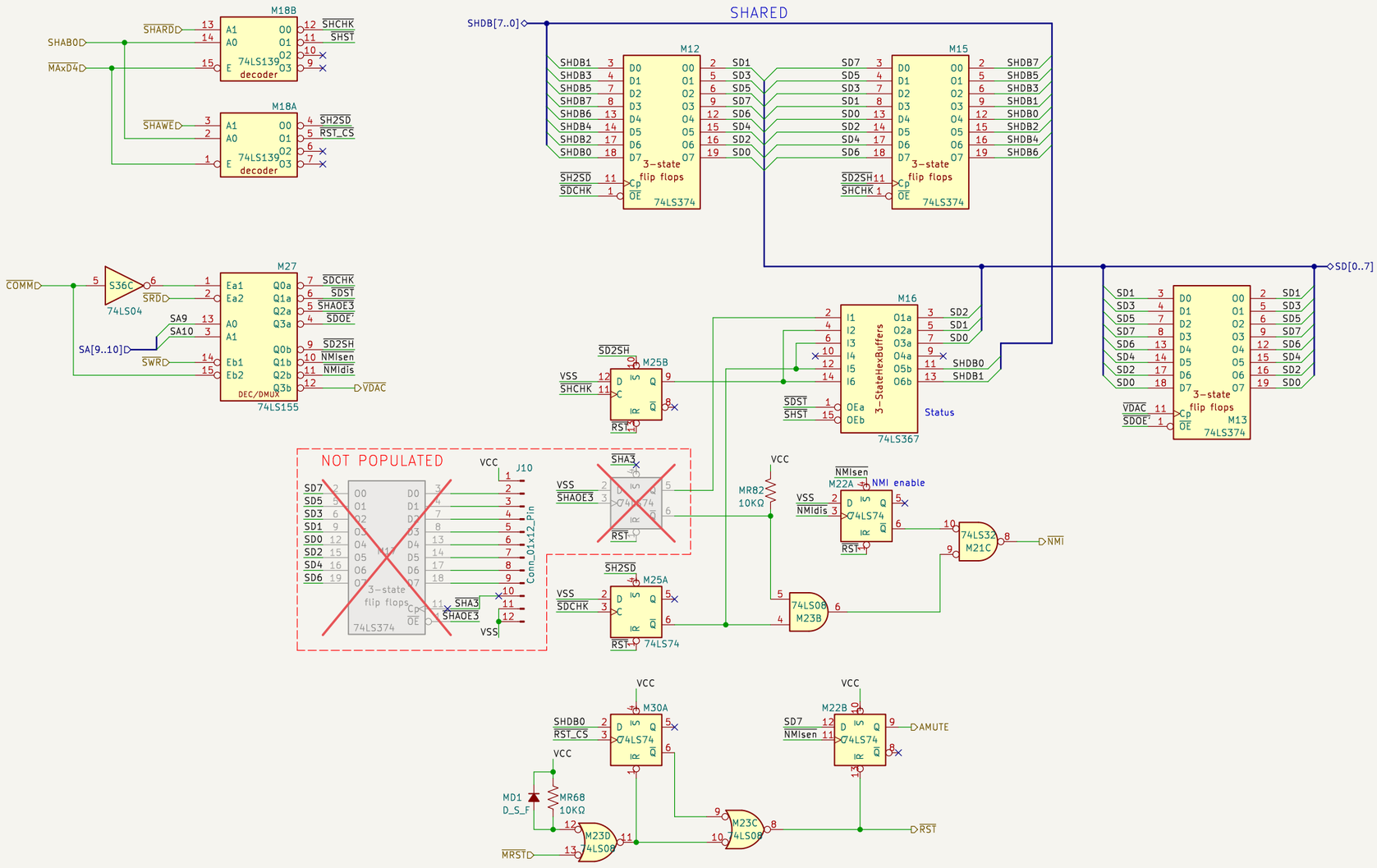


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Sheet: /Video/Line buffer/
File: linebuf.kicad_sch

Title: Single Line Video Buffer

Size: A3	Date: 2024-11-16	**Rev:**
KiCad E.D.A. 8.0.5		Id: 17/20

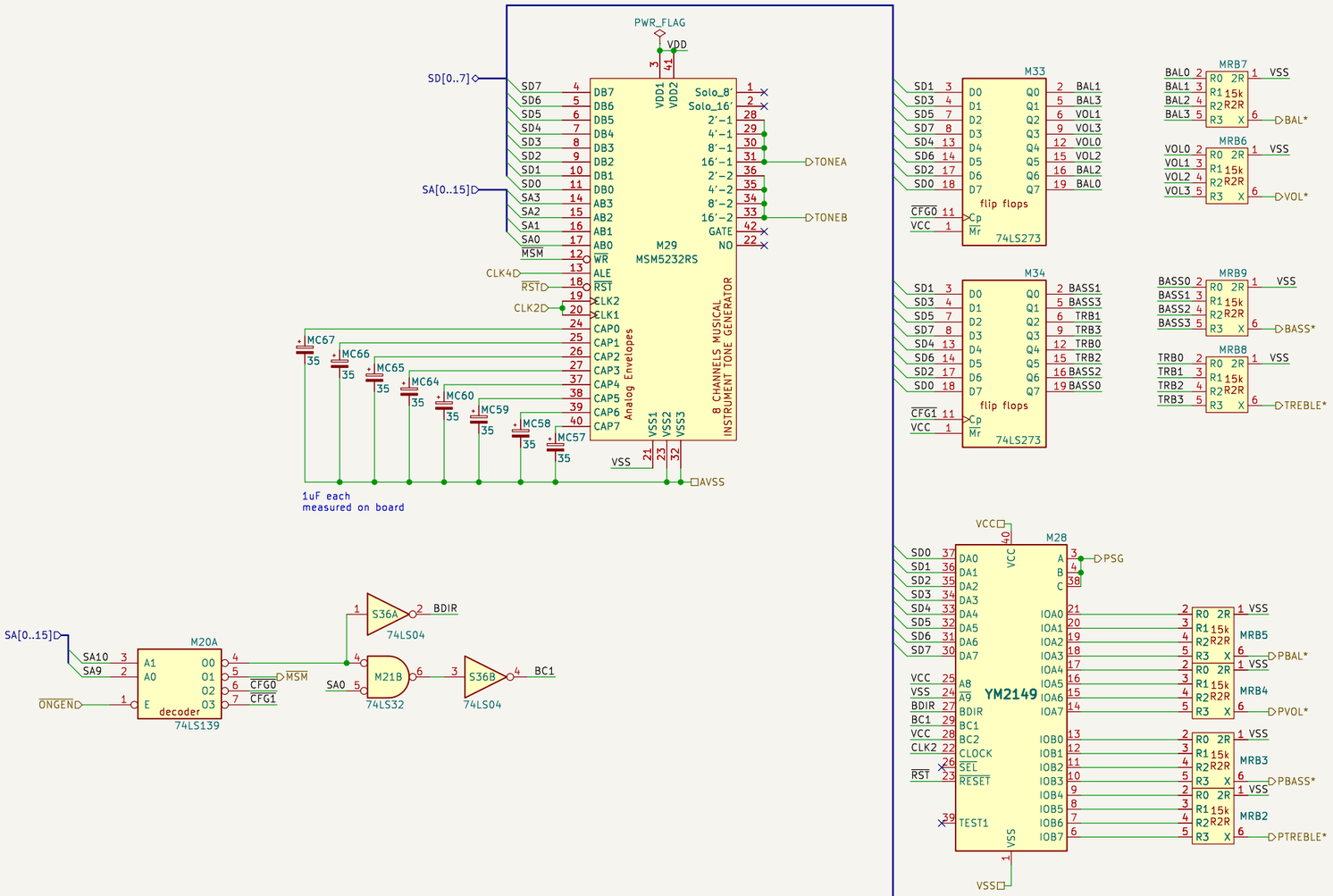


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Sheet: /Sound/Latches/
File: latches.kicad_sch

Title: Communication Latches

Size: A3	Date: 2024-11-16	**Rev:**
KiCad E.D.A. 8.0.5		Id: 18/20



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Sheet: /Sound/Sound Generators/
File: ongen.kicad_sch

Title: Sound Generators

Size: A3	Date: 2024-11-16	**Rev:**
KiCad E.D.A. 8.0.5		Id: 19/20

Access to sound generators
comes with one wait state

MSM5232 address latch
is triggered automatically
when a write occurs

SR78
2.2KΩ

SR79
5.6KΩ

SR77
1.8KΩ

SR80
2.2KΩ

SC62
C

TP5

SX1 8MHz

M35A
M35B
M36E
M36D
M36F
M26B
M26A
M23A
74LS08
M32A
M32B
M31A
S31B
M30B
M21D
74LS32
MJP1
Jumper_4

74LS74
74LS04
74LS393
BINARY COUNTER

VCC
VSS

CLK4
CLK2
SCLK
WAIT
INT
MSM
SCE8
M1
IORQ

4MHz
250kHz
15.625kHz
976.6Hz
122Hz

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Sheet: /Sound/Clocks/
File: snd_clocks.kicad_sch

Title: Sound Clock Generation

Size: A3	Date: 2024-11-16	**Rev:**
KiCad E.D.A. 8.0.5		Id: 20/20